

Menus & how to open them

Edge menu edit mesh

Vertex menu edit mesh

Face menu edit mesh

Context menu edit / object

Object menu object

Ctrl + E

Ctrl + V

Ctrl + F

Right-click

Object ▼ header

CUTTING & BOOLEANS

01

Hard Ops

The hub of the hard-surface workflow — booleans, bevels, mirroring, modifier management, and material/viewport helpers, all behind one menu.

WHEN Any destructive/boolean hard-surface work; managing the modifier stack.

main menu

pie menu

HOps Helper

materials

viewport

interactive mirror

bevel helpers

Q

Shift + Q

Ctrl + `

Alt + M

Alt + V

Alt + X

Ctrl + Shift + B

Booleans (object mode):

Ctrl + Num - difference · + union · *

intersect · ÷ slash.

BoxCutter

Draw-on-surface cutting. Sketch a shape and it becomes a live boolean cut, inset, or new object. Pairs with Hard Ops.

WHEN Fast carving of panels, slots, and shapes directly in the viewport.

activate BoxCutter

Alt + W

Tool active:

Ctrl + D pie menu · D helper · C custom.

PUNCHit

MACHIN3, DeusEx bundle

Manifold inset + extrude — "punch" clean holes and recesses that stay watertight, without the boolean cleanup.

WHEN Insetting/extruding faces where you need manifold, fuse-ready geometry.

edit mesh

Shift + E

SURFACE REFINEMENT

02

MESHmachine

Post-bevel surgery: rework chamfers and fillets after the fact, fuse/unfuse/refuse surfaces, flatten, normal-edit, and stamp on detail with the Plug library.

WHEN Fixing or reshaping bevels/fillets that a normal bevel can't handle.

menu edit + object

Y

All tools live inside this one menu.

Edge Flow

MENU-ONLY

Relaxes selected edges/loops to follow surrounding surface curvature (like Maya's Edit Edge Flow). Also sets edge/vertex curve.

WHEN Cleaning up edge loops so they flow smoothly across curved surfaces.

Edge menu → Set Flow / Curve / Linear

Ctrl + E

Vertex menu → Set Vertex Curve

Ctrl + V

Also in the edit-mode

Right-click context menu → Set Flow.

CURVES

03

CURVEmachine

MESHmachine's little brother for curves — building and editing clean curve geometry for hard-surface detailing.

WHEN Cable runs, pipes, and curve-based detail work.

menu curve edit mode

Y

Curves to Mesh

Generates a clean quad surface that flows between two or more curves.

WHEN Building surfaces/patches from curve "rails."

run from the curve tools

While live:

Alt + ↑↔→ subdivisions · Ctrl + R refresh · Shift + V preview · Alt + Enter commit.

CAD-STYLE PRECISION

04

Construction Lines

CAD-style accurate modelling — work to exact dimensions with construction guides, snapping, and numeric input.

WHEN Anything that needs real-world measurements and precise constraints.

launch key left of 1

Alt + `

Then follow the on-screen prompts.

Grid Modeler

MENU-ONLY

Draw shapes on an interactive grid to cut into or build out geometry, with precise grid snapping.

WHEN Greebles, panel detailing, and precise on-surface shapes.

Edit →

Right-click → Grid Modeler

While live:

Q / Enter confirm · Ctrl + Scroll resolution · Alt + Scroll extend grid · W / RMB switch mode · Z close shape · X from edge.

nerdz.cloud

Mesh Align Plus

Align, move, and rotate geometry using points/lines/planes picked from the mesh itself — precise alignment without eyeballing.

WHEN Matching parts together exactly using existing geometry as reference.

Sidebar → Align Plus tab

N

Simple Bend MENU-ONLY

Bend geometry along an axis interactively, with angle/segment control.

WHEN Curving a straight object (pipes, brackets, signage) to a set angle.

Edit → **Right-click** → Simple Bend

While live: **W** swap axis · **C** one side · **S** selected verts · **Shift** snap 5° · **Q** confirm.

WORKFLOW & UTILITY

05

MACHIN3tools

A large suite of quality-of-life tools and pie menus: Smart Vert/Edge/Face, Focus, Mirror, Group, customised pies, and Hard Ops / BoxCutter integration.

WHEN Everyday modelling speed-ups; ties the other tools together.

modes pie

Tab

focus **F** / **Ctrl** + **F**

group **Ctrl** + **G**

symmetrize **Alt** + **X**

edge-constrained transform **Alt** + **R**

smart select **Alt** + **LMB**

Quick Snap

Maya-style interactive vertex/point snapping — grab a point and snap it precisely onto another.

WHEN Precise point-to-point placement during edit/object mode.

snap

Ctrl + **Shift** + **V**

Select Sim MENU-ONLY

Quickly select faces/edges similar to the current selection (by shape hash).

WHEN Grabbing all matching faces/edges in one step.

Edit → **Right-click** → Select Sim

Conform Object MENU-ONLY

Wrap / conform one object onto the surface of another.

WHEN Laying detail, decals-as-geometry, or panels onto a curved surface.

Object → **Object** ▾ or **Right-click** → Conform Object

Mesh Copier MENU-ONLY

Copy mesh detail from a source edge-ring/face onto destination faces.

WHEN Reusing mesh detail across multiple surfaces.

Edit → **Right-click** → Mesh Copier

Set Source, then Paste.

DETAILING ASSETS

06

Just Panels

Generates panelling/greeble detail, with an accompanying asset pack.

WHEN Fast sci-fi / hard-surface surface detailing.

pie menu **Alt** + **K**

Install the addon zip; load **JustPanels_ASSETS** via Preferences → File Paths → Asset Libraries.

Shortcuts reflect the addons' default keymaps as verified in Blender 5.0 (no personal overrides). "Menu-only" tools register no keymap of their own — they live solely inside the listed menus.